LAKEVIEW UNIVERSITY

Department of Mathematics
MATH 231 - Discrete Mathematics
Spring 2026

QUICK REFERENCE HANDOUT: Logic, Proofs, and Set Notation
Prepared by the course staff. Keep this sheet for the semester.

PART 1: PROPOSITIONAL LOGIC

CONNECTIVES AND TRUTH VALUES

Symbol   Name   Read as
P, Q, R   Propositions   Variables representing true/false statements
!P   Negation   "not P"
P & Q   Conjunction   "P and Q"   True only when both are true
P | Q   Disjunction   "P or Q"   True when at least one is true
P -> Q   Conditional   "if P then Q"   False only when P true, Q false
P <-> Q   Biconditional   "P if and only if Q" True when both same value

CONDITIONAL FORMS

Original: P -> Q "if P then Q"
Converse: Q -> P "if Q then P" (NOT equivalent to original)
Inverse: !P -> !Q "if not P then not Q" (NOT equivalent to original)
Contrapositive: !Q -> !P "if not Q then not P" (EQUIVALENT to original)

KEY LOGICAL EQUIVALENCES

Double negation: !!P = P
De Morgan 1: !(P & Q) = !P | !Q
De Morgan 2: !(P | Q) = !P & !Q
Implication rewrite: P -> Q = !P | Q
Contrapositive: P -> Q = !Q -> !P

PART 2: PROOF TECHNIQUES

DIRECT PROOF

Goal: Prove P -> Q
Method: Assume P is true. Use definitions, theorems, and algebra to derive that Q must also be true.
When to use: When a clear chain of reasoning connects P to Q.

PROOF BY CONTRAPOSITIVE

Goal: Prove P -> Q

Method: Prove the contrapositive: !Q -> !P

Assume Q is false. Derive that P must be false.

When to use: When it is easier to work backwards from Q being false.

PROOF BY CONTRADICTION

Goal: Prove statement S

Method: Assume S is false (i.e., !S). Derive a logical contradiction (something that is always false, or two statements that cannot both be true). Conclude that !S cannot hold, so S must be true.

When to use: Existential claims, irrationality proofs, and when assuming false leads naturally to a contradiction.

Classic example: Proof that sqrt(2) is irrational.

PROOF BY INDUCTION

Goal: Prove a statement P(n) is true for all integers n >= base case

Steps:

1. Base case: Verify P(base) is true (usually P(0) or P(1))
2. Inductive step: Assume P(k) is true (the inductive hypothesis). Use that assumption to prove P(k+1) is also true.

When to use: Statements about natural numbers, sums, inequalities, properties of sequences.

Important: The inductive hypothesis only lets you assume P(k). You MUST prove P(k+1) from it.

STRONG INDUCTION

Same as induction but the inductive hypothesis is: assume P(j) is true for all j where base <= j <= k. Useful when P(k+1) depends on more than just the immediately preceding case.

PART 3: SET NOTATION AND OPERATIONS

BASIC NOTATION

{a, b, c} Set with listed elements

x in A x is an element of set A

x not in A x is not an element of A

A is a subset of B Every element of A is also in B (A can equal B)

A is a proper subset of B A is a subset of B and A != B

Empty set The set with no elements (written as empty braces or O/)

|A| **Cardinality:** the number of elements in A

COMMON NUMBER SETS

N Natural numbers: {0, 1, 2, 3, ...} (some definitions start at 1)

Z Integers: {..., -2, -1, 0, 1, 2, ...}

Q Rational numbers (fractions p/q where p, q are integers, q != 0)

R Real numbers

Z+ Positive integers: {1, 2, 3, ...}

SET OPERATIONS

A union B Elements in A or B or both

A intersect B Elements in both A and B

A minus B Elements in A but not in B

Complement of A All elements (in the universal set) not in A

A x B **Cartesian product:** all ordered pairs (a, b) with a in A, b in B

Power set of A The set of all subsets of A; if |A| = n, |P(A)| = 2^n

SET IDENTITIES (useful for proofs)

A union empty = A

A intersect empty = empty

A union A = A

A intersect A = A

A union (B intersect C) = (A union B) intersect (A union C) [distributive]

Complement of (A union B) = Complement(A) intersect Complement(B) [De Morgan]

Complement of (A intersect B) = Complement(A) union Complement(B) [De Morgan]

NOTES FOR EXAMS

- Show all steps in proofs. An answer with no justification receives no credit.
- For induction proofs, clearly label the base case and the inductive step.

- When asked to disprove a statement, one counterexample is sufficient.
- **Set equality:** to prove A = B, prove A is a subset of B AND B is a subset of A.
- Carefully distinguish between "element of" and "subset of" in your answers.

Questions? Bring them to office hours or post on the course portal discussion board.